Lines and Planes

Questions 1–25

Question 1. Find the acute angle between the line with equation $\mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ -9 \end{pmatrix}$ and each of the following planes:

a) $x = 0$
b) $y = 0$
c) $z = 0$

Question 2. Find the acute angle between the line with equation $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix}$ and the plane with equation $4x + y - 2z = 3$.

Question 3. Determine the geometrical configuration of the line with equation $\mathbf{r} = \begin{pmatrix} 3 \\ 7 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -8 \\ 9 \end{pmatrix}$ and the plane $6x + 15y + 10z - 90 = 0$.

Question 4. Determine the geometrical configuration of the line with equation $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ \frac{11}{3} \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -4 \\ -5 \end{pmatrix}$ and the plane $3x + 12y - 6z + 13 = 0$.

Question 5. Determine the geometrical configuration of the line with equation $\mathbf{r} = \begin{pmatrix} -1 \\ 5 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix}$ and the plane $5x + 3y + 2z - 4 = 0$.

Question 6. A road can be modelled as a plane using the equation $x + y + 10z = 1$, where x, y and z are measured in metres. A gas pipe can be modelled as a straight line passing through the points $(1, 1, -0.5)$ and $(0.2, 0.4, -0.6)$.

a) Find the acute angle that the line makes with the plane.

An engineer must connect a monitoring device to the pipe from a point on the road, P. It is given that P has coordinates $(0.8, 92.1, -9.19)$.

a) Find the shortest distance from the point P to the pipe.

Question 7. Find the coordinates of where the line with equation $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 1 \\ -2 \end{pmatrix}$ intersects the following planes:

a) $x = 0$
b) $y = 0$
c) $z = 0$

Question 8. Find the coordinates of where the line with equation $\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix}$ intersects the plane with equation $4x + y - z = 9$.

Question 9. Point A has coordinates $(3, 2, 4)$. Find the exact coordinates of the image of point A when it is reflected in the plane with equation $x + y - z = 0$.

Question 10. Point P has coordinates $(5, -2, \frac{1}{2})$. Find the exact coordinates of the image of point P when it is reflected in the plane with equation $10x - y - 2z = -54$.

Question 11. It is given that l_1 intersects the plane Π at the point P, and l_2 intersects the plane Π at the point Q, where

$$l_1 : \ \mathbf{r} = \begin{pmatrix} 2 \\ 8 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix}, \qquad l_2 : \ \mathbf{r} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \qquad \Pi : 9x - 20y + 2z = 6.$$

Find the exact length of PQ.

Question 12. The line l has the equation $\mathbf{r} = \begin{pmatrix} 2 \\ a \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} b \\ 4 \\ 1 \end{pmatrix}$ and the plane Π has the equation $2x + y + z = 5$.

a) Given that the line lies on the plane Π, find the values of a and b.

b) Given instead that the line l intersects the plane Π at precisely one point, state the condition on a and b for this to be the case.

Question 13. The line l has equation $\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$ and the point P has coordinates $(-3, 7, 5)$.

a) Verify that P lies on the line l.

It is given that the plane Π is perpendicular to the line l and contains the point P.

a) Find the cartesian equation of the plane Π.

The plane Π contains the point $Q\left(-3, 7 - \frac{1}{\sqrt{5}}, 5 + \frac{2}{\sqrt{5}}\right)$. The point $R(a, b, c)$ lies on the line l such that the distance QR is 5 units.

a) Find the possible coordinates of R.

Question 14. Points A and B have coordinates $(3, -2, 3)$ and $(4, -1, -2)$ respectively and lie on the line l_1.

a) Find the equation of l_1 in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$.

The points A and B lie on the plane Π with equation $px + qy + rz + d = 0$, where p, q, r and d are constants. The point C with coordinates $(1, 2, 1)$ also lies on Π.

a) Find the equation of l_2, the line that passes through A and C, in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$.

b) Show that $\mathbf{n} = 3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ is perpendicular to both lines l_1 and l_2.

c) Hence find the equation of the plane Π.

Points D and E have coordinates $(8, 5, 9)$ and $(6, 4, 10)$ respectively and lie on the line l_3. The line l_3 intersects the plane Π at the point P.

a) Find the exact coordinates of the point P.

b) Find the acute angle that is made when l_3 intersects the plane Π.

It is given that the point F lies on l_3 such that $PF = 1$ unit.

a) Find the possible coordinates of F.

Question 15. Plane Π has equation $\mathbf{r} \cdot \begin{pmatrix} 3 \\ -9 \\ 2 \end{pmatrix} = 18$. Determine whether the following lines intersect at a single point, or whether they lie within Π, or whether they are parallel to Π. If they intersect the plane, determine the coordinates of this point of intersection:

a) $\mathbf{r} = \begin{pmatrix} 2 \\ 5 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$

b) $\mathbf{r} = \begin{pmatrix} 6 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 13 \\ 48 \end{pmatrix}$

c) $\frac{x-6}{2} = \frac{y-4}{4} = \frac{z-18}{15}$

d) $x + 8 = 3 - y = \frac{z+1}{6}$

Question 16. Two lines l_1 and l_2 are described as follows:

$$l_1 : \ \frac{x+5}{3} = y - 1 = 2 - z, \qquad l_2 : \ 5 - x = y + 3 = \frac{z-5}{2}.$$

a) Show that l_1 and l_2 are skew lines.

b) Find a vector that is perpendicular to both l_1 and l_2.

c) Find a cartesian equation of the plane which is parallel to l_1 and contains l_2.

d) Find a cartesian equation of the plane which is parallel to l_2 and contains l_1.

The line l_3 has equation $\mathbf{r} = \begin{pmatrix} 2 \\ 5 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 3 \\ 8 \end{pmatrix}$. The line l_3 intersects the plane found in (c) at the point P and the plane found in (d) at the point Q.

a) Find the exact distance PQ.

Question 17. Find the coordinates of where the line with equation $2x + 8 = \frac{4-y}{3}$, $z = 2$ intersects the plane with equation $\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 7 \\ 3 \\ 5 \end{pmatrix}$.

Question 18. One face of a mountainside can be modelled by a plane that passes through the points $(60, 70, 320)$, $(50, 120, 235)$ and $(130, 100, 155)$, where measurements are in metres.

a) Find a vector equation of the plane in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$.
b) Show that the vector $3\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$ is perpendicular to the plane.
c) Hence write down the cartesian equation of the plane.
d) A tunnel is to be dug into the mountainside that is perpendicular to the rock face, so that it passes through the point $(20, 65, 100)$. Find the coordinates of the point where the tunnel should begin and find the exact length of the tunnel so that the length of the tunnel is minimised.

Question 19. It is given that

$$\Pi_1 : 4x - y - z = 3, \qquad \Pi_2 : 3x + 2y + 2z = 9.$$

a) Show that Π_1 and Π_2 are not perpendicular.
b) Find a point that lies on both Π_1 and Π_2.
c) Find a vector that is parallel to both Π_1 and Π_2.

It is given that the plane Π_3 is perpendicular to both Π_1 and Π_2.

a) Find the equation of the plane Π_3 that passes through the point $(5, 3, 1)$.

Question 20. It is given that

$$\Pi : 5x + 8y + 9z = 16, \qquad l : \ \mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix}.$$

The points P and Q lie on l so that they are both 14 units from Π.

a) Find the coordinates of P and Q.

It is given that the point A has coordinates $(11, -6, 1)$ and that A lies on Π.

a) Find the exact area of the triangle APQ.

Question 21. It is given that

$$\Pi : x - y + z = 6, \qquad l : \ \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -7 \\ 6 \end{pmatrix}.$$

Find an equation of the line that is the reflection of the line l in the plane Π.

Question 22. It is given that

$$\Pi_1 : 6x + y - 2z = 8, \qquad \Pi_2 : 5x - 3y - 10z = 2.$$

Show that the line l with equation

$$\mathbf{r} = \begin{pmatrix} 22 \\ -64 \\ 30 \end{pmatrix} + \lambda \begin{pmatrix} 16 \\ -50 \\ 23 \end{pmatrix}$$

lies on both planes.

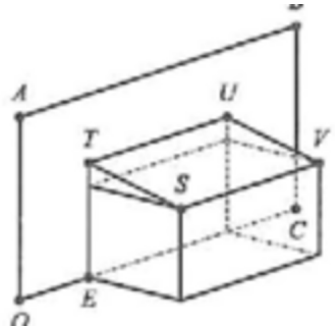
Question 23. Find the equation of the line of intersection between the planes:

$$\Pi_1 : 2x - y + z = 5, \qquad \Pi_2 : 3x + y - z = 2.$$

Question 24. Find the equation of the line of intersection between the planes:

$$\Pi_1 : z = 0, \qquad \Pi_2 : 5x + 2y + 3z = 8.$$

Question 25. The diagram shows a rabbit hutch built against a wall.



The wall $OABC$ can be modelled by the plane with equation $x = 0$. The rabbit hutch $STUV$ can be modelled by the plane with equation $x - 4z + 20 = 0$.

a) The lid of the rabbit hutch is in contact with the wall along the line TU. Find the equation of the line passing through T and U in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$.

b) Find the acute angle ETS.